

Connecting Power BI to SAP Datasphere — architecture options and implementation plan

Version 0.1 — discussion draft. Assumptions marked [VERIFY] must be confirmed before this becomes a commitment.

1. Framing the problem correctly

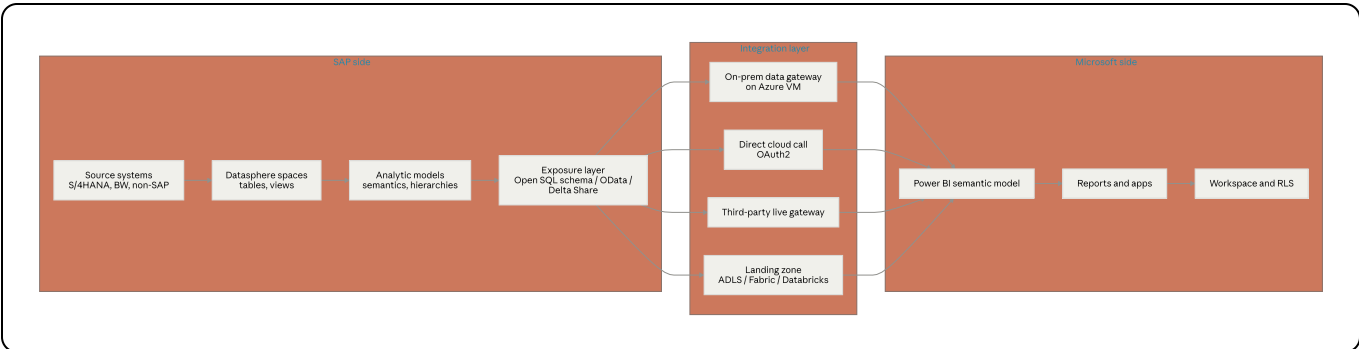
The question "how do we connect Power BI to Datasphere" has five technically valid answers. Choosing between them is not a connectivity question — it is a decision about **where the semantic layer lives and who enforces security**.

Three questions determine the answer. Everything else is detail.

Question	If answer A	If answer B
Where do business semantics live? (hierarchies, currency conversion, calculated measures)	Datasphere → you need a path that preserves them	Power BI → you can pull flat tables and rebuild
Who enforces row-level security?	Datasphere (Data Access Controls) → you need per-user identity propagation	Power BI (DAX RLS) → a technical service user is acceptable
What is the data volume and latency requirement?	Large / near-real-time → federation or replication	Modest / daily → import works fine

If the customer cannot answer these, the deliverable is a **decision workshop**, not an architecture. Say so.

2. Reference architecture — the layers involved



The single most important design point sits at the boundary: **an analytic model in Datasphere and a semantic model in Power BI do the same job**. Building both means

maintaining business logic twice, and they will diverge. Decide which one is authoritative before writing a line of M code.

3. The five connectivity options

Option 1 — Open SQL schema via SAP HANA ODBC

Datasphere exposes a space's objects to a database user. Power BI connects with the SAP HANA ODBC driver (or the generic ODBC connector).

How it works

1. In Datasphere: Space Management → Database Access → create a database user with read access.
2. Add the connecting client's public IPv4 address to the Datasphere IP allowlist.
3. Install the SAP HANA client / ODBC driver on the machine that will connect — the developer's desktop for authoring, and the gateway VM for scheduled refresh.
4. Power BI Desktop → Get Data → SAP HANA database or ODBC → Import or DirectQuery.
5. For the Power BI Service, register the gateway data source and bind the semantic model to it.

Aspect	Assessment
Performance	Best of all options. Native SQL pushdown; HANA does the work.
DirectQuery	Supported
Semantics preserved	No. You see the exposed views as flat relational objects. Technical field names, no hierarchies, no analytic-model measures.
Security	Connects as a database user , not the report viewer. Datasphere sees one identity. RLS must be rebuilt in Power BI DAX, or SSO must be configured.
Infrastructure	Gateway VM required for the Service. Static outbound IP required for the allowlist.
Operational burden	Medium — driver version management, password rotation policy on the database user, IP allowlist maintenance.

The governance objection you will hear: "that database user can see everything in the space." It is a fair objection. Answers: scope one database user per space with a minimal set of exposed views; or configure SAML/Kerberos SSO through the gateway so Datasphere applies the real user's privileges. SSO is materially harder — plan a week for it, not a day.

Option 2 — OData

Datasphere exposes analytic models or relational views as OData services. Power BI consumes them with the OData or Web connector over OAuth2.

Two flavors:

- **Analytical OData** — consumes an analytic model, preserving aggregation behavior and business semantics.
- **Relational OData** — consumes fact and dimension views as flat entities.

Aspect	Assessment
Performance	Weakest. HTTP pagination; poor for large extracts.
DirectQuery	Not supported — import only.
Semantics preserved	Partially, and better than ODBC when using the analytical flavor.
Security	OAuth2 client credentials or authorization code flow. Token refresh in the Power BI Service is the recurring pain point.
Infrastructure	None — cloud to cloud, no gateway, no driver, no VM.
Operational burden	Low to set up, medium to keep running.

When this is the right answer: modest volumes, a small number of curated analytic models, and an organization that does not want to run a gateway VM. It is by far the fastest route to a working proof of concept — use it in week 1 to demonstrate feasibility even if it is not the final architecture.

Option 3 — Third-party live connectivity gateway

Commercial products (APOS Live Data Gateway, CData drivers, Theobald) sit between the two platforms and translate the SAP semantic layer into something Power BI can consume live.

Aspect	Assessment
Performance	Good; live query
DirectQuery / Live	Supported
Semantics preserved	Yes — this is the entire value proposition. Hierarchies, currency conversion, and Datasphere-defined measures survive.
Security	Vendor-dependent; most support identity propagation so Datasphere DACs apply per user
Infrastructure	Vendor server component to install and operate
Operational burden	Medium, plus vendor dependency
Cost	License fee ([VERIFY – request quotes])

When this is the right answer: the customer has invested heavily in Datasphere analytic models and refuses to rebuild that logic in DAX, and per-user security must be enforced by Datasphere. This is the option that solves both hard problems at once, and it is the one procurement will resist.

Option 4 — Replicate out to a Microsoft landing zone

Datasphere replication flows (or a scheduled pull through the Open SQL schema) push data into Azure Data Lake Storage, Azure SQL, a Fabric Lakehouse, or Databricks. Power BI reads from there.

Aspect	Assessment
Performance	Best at scale; refresh is decoupled from SAP
Mode	Import, or Direct Lake if landing in Fabric
Semantics preserved	No — rebuilt on the Microsoft side
Security	Fully Microsoft-side
Infrastructure	A full data platform: storage, orchestration, monitoring
Operational burden	High — this is a data engineering program, not a connection
Cost	Storage and compute, plus Datasphere outbound integration charges [VERIFY: Premium Outbound Integration blocks]

When this is the right answer: high volumes, many downstream consumers beyond Power BI, or an existing Fabric/Databricks investment. Given a Databricks-centric stack, this option and option 5 converge.

Option 5 — BDC Connect / Delta Sharing (the strategic path)

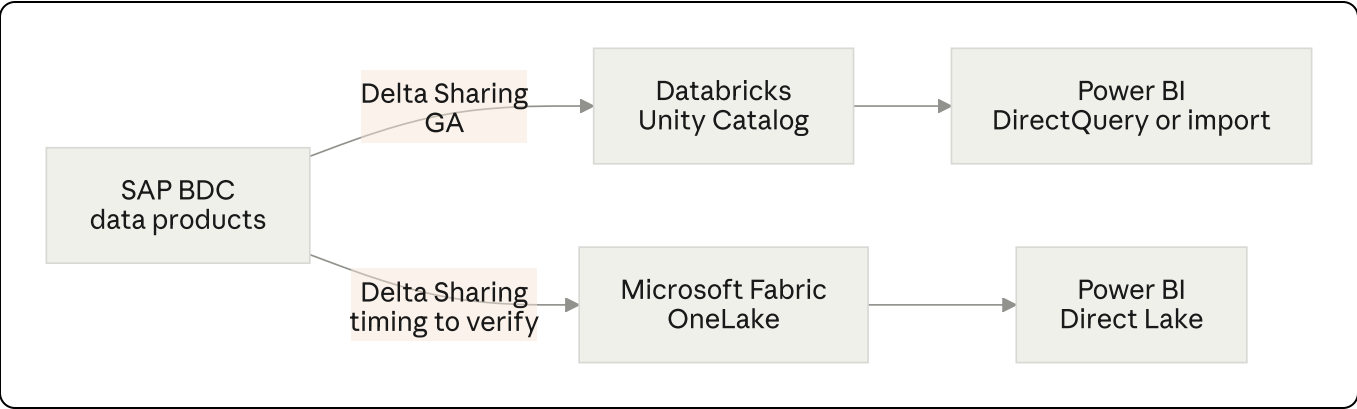
SAP Business Data Cloud implements the open Delta Sharing protocol over its object store, allowing zero-copy bidirectional sharing of SAP data products.

Current status:

- **Databricks:** generally available since October 2025.
- **Google BigQuery and Snowflake:** first half of 2026.
- **Microsoft Fabric:** announced for Q3 2026 — [VERIFY current GA status before presenting].

Aspect	Assessment
Performance	Zero-copy federated access; no pipeline latency
Semantics preserved	Yes — SAP business context travels with the data product
Security	Governed on both sides; Unity Catalog on the Databricks side
Infrastructure	Requires SAP Business Data Cloud , not plain Datasphere
Operational burden	Low once established — click-driven, no ETL code
Availability risk	The Fabric path is the one the customer wants and the one whose date is least certain

Two routings today:

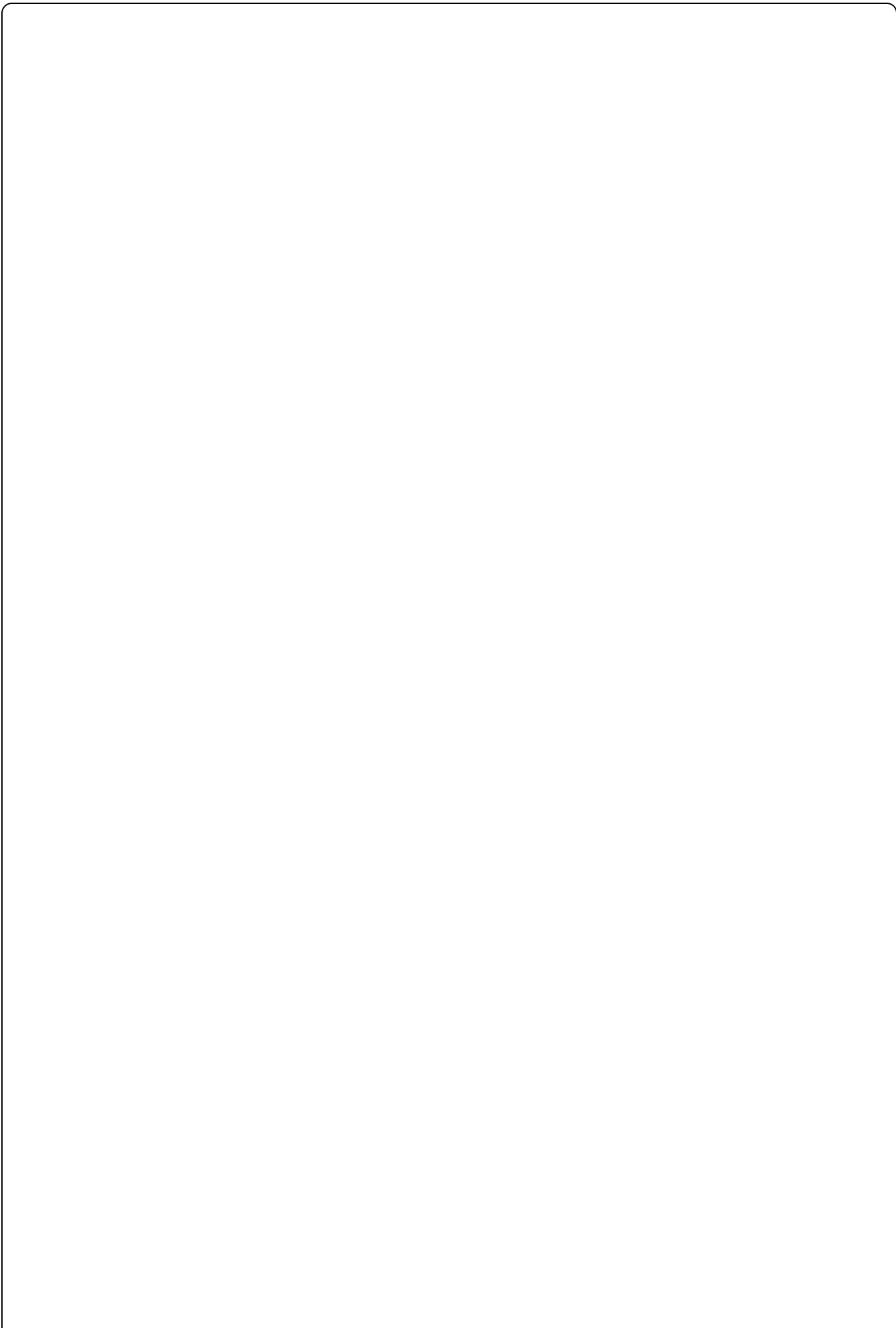


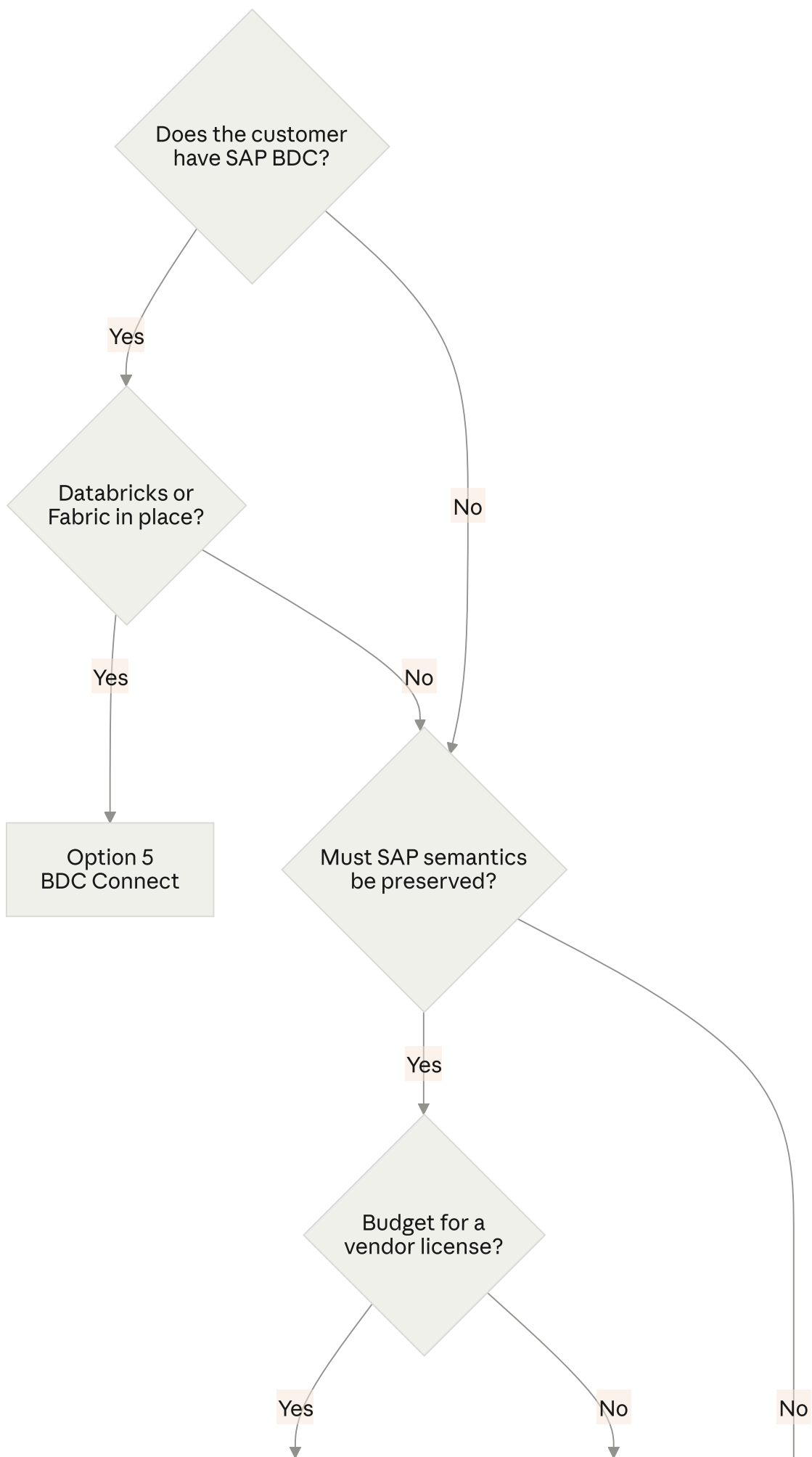
If the customer already runs Databricks, the available path today is **BDC → Databricks → Power BI**, which works now and positions them for the native Fabric path when it lands. That is the recommendation worth making in the architecture document, with the caveat that it depends on BDC licensing.

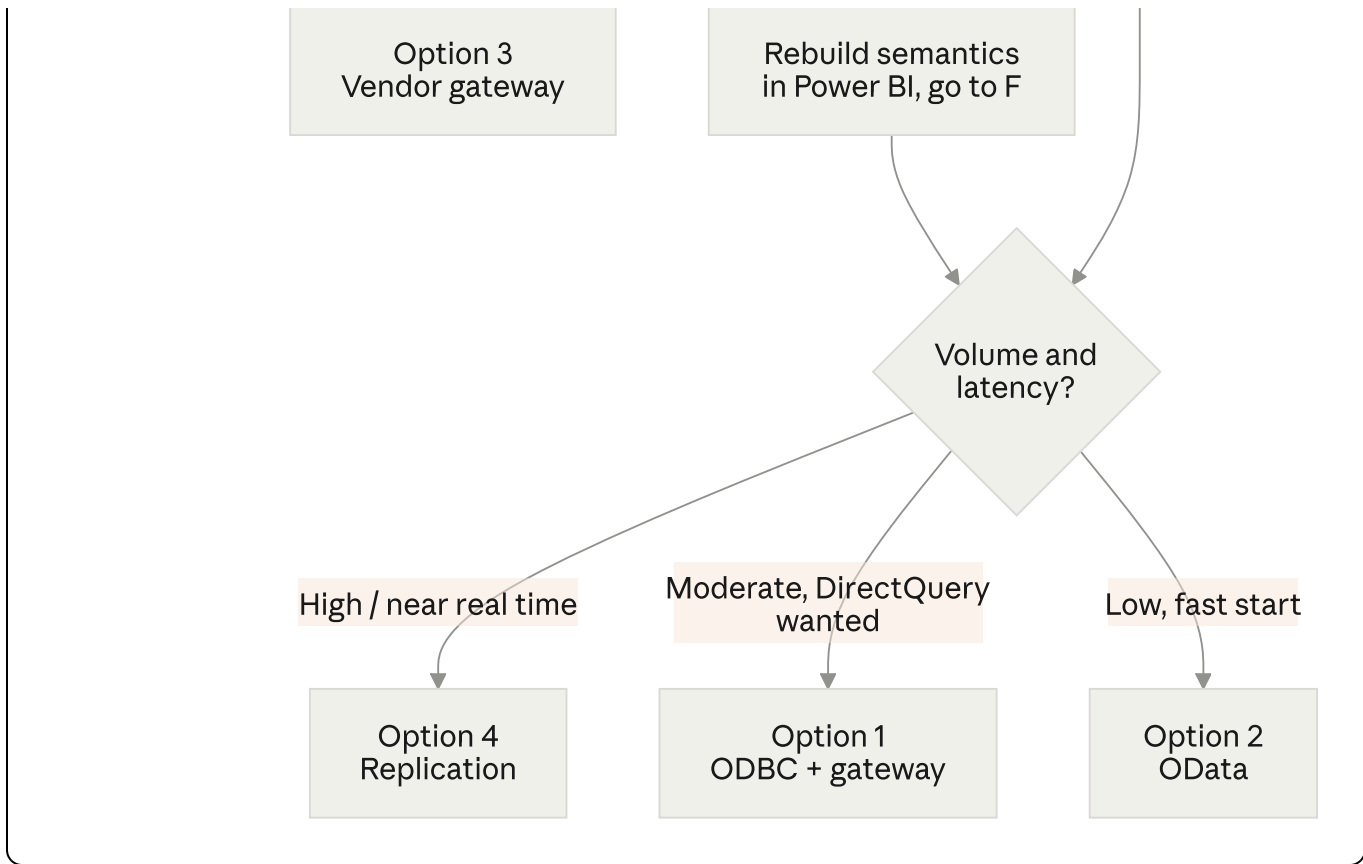
4. Comparison matrix

	1. ODBC	2. OData	3. Vendor gateway	4. Replication	5. BDC Connect
Setup effort	Medium	Low	Medium	High	Medium
Time to first report	1 week	2 days	2 weeks	4+ weeks	3+ weeks
Performance at scale	High	Low	High	Highest	High
DirectQuery / live	Yes	No	Yes	Direct Lake	Yes
Preserves SAP semantics	No	Partial	Yes	No	Yes
Per-user security in SAP	Only with SSO	Possible	Yes	No	Governed
Gateway / VM needed	Yes	No	Yes	Yes	No
Additional license cost	No	No	Yes	Infra cost	BDC required
Data duplication	Optional	Yes	No	Yes	No
Strategic alignment	Tactical	Tactical	Tactical	Medium	High

Decision tree







5. Security design — the part that gets under-designed

This is where these projects fail review. Three models, pick one deliberately:

Model	How it works	Consequence
Technical user + Power BI RLS	One Datasphere database user; all security expressed in DAX roles	Simplest. Datasphere Data Access Controls are bypassed entirely. Security is duplicated and can drift.
SSO with identity propagation	SAML or Kerberos through the gateway; Datasphere applies the real user's analytic privileges	Correct. Significantly harder: identity mapping (UPN to Datasphere external identity, case-sensitive), certificate management, encryption must be enabled before establishing SAML SSO on the HANA path.
Vendor gateway with propagation	Product handles identity pass-through	Correct and easier, at a license cost

Recommendation for the document: state plainly that if the customer's Data Access Controls in Datasphere are load-bearing for compliance, the technical-user model is not acceptable, and the cost of the alternatives should be evaluated up front rather than discovered in UAT.

Additional security items to cover:

- Database user password rotation policy and its effect on unattended refresh
- IP allowlist management and the need for a static outbound IP on the gateway VM
- Where sensitivity labels are applied (Purview on the Microsoft side has no visibility into Datasphere classifications)
- Audit trail: Datasphere logs the database user, Power BI logs the report viewer — correlating them requires deliberate design

6. Proof of concept plan

Do not choose on paper. Two to three weeks of testing produces evidence the customer can act on.

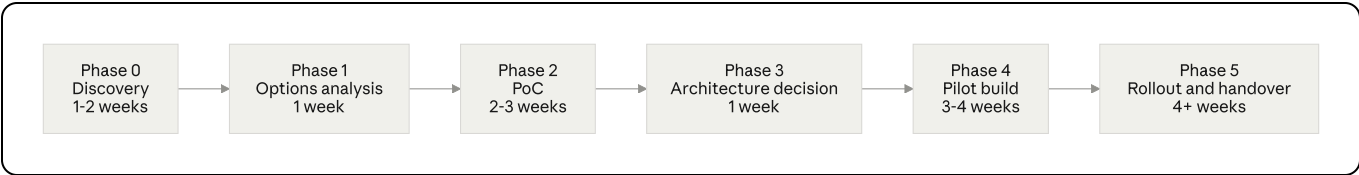
Week	Activity	Output
1	Provision access: Datasphere database user, IP allowlist, gateway VM, test analytic model	Working Option 2 (OData) connection — proves feasibility fast
2	Build the same report on Option 1 (ODBC, import and DirectQuery)	Performance comparison on identical logic
3	Test the strategic option available to the customer (3, 4, or 5) + security test	Recommendation with measured evidence

Use **one representative model** for all tests — ideally the largest one the business actually cares about, not a toy table.

Measure these, and put the numbers in the deck

Metric	Why
Initial load time and row count	Baseline
Incremental refresh duration	Determines whether the refresh window is viable
DirectQuery visual response time (p50, p95)	The number users will judge you on
Hierarchies and currency conversion: preserved or lost	Determines rework effort
Datasphere capacity units consumed per refresh	Cost — often the surprise
Behavior when a security-relevant user changes	Proves the security model works

7. Implementation plan



Phase	Duration	Deliverable	Difficulty
0. Discovery	1–2 weeks	Current-state landscape, requirements, volume and latency profile, security requirements	Low
1. Options analysis	1 week	This document, customer-specific, with scoring against weighted criteria	Low
2. Proof of concept	2–3 weeks	Measured comparison of 2–3 options	Medium
3. Architecture decision	1 week	Signed-off target architecture, ADR log, cost model	Low, high stakes
4. Pilot build	3–4 weeks	One production-grade model end to end, with security and refresh	Medium–high
5. Rollout	4+ weeks	Remaining models, monitoring, documentation, handover	Medium

Phases 0–3 total: 5–7 weeks. That is the scope of the current request. Phases 4–5 should be quoted separately.

Resources and dependencies

Resource	Role	Availability needed
Data architect (this role)	Design, PoC, documentation	Full time phases 0-3
SAP Datasphere administrator	Space access, database users, IP allowlist, exposure of test models	Critical dependency — 20%, but responsive
Power BI / Fabric administrator	Gateway registration, tenant settings, workspace	10%
Azure / infrastructure	Gateway VM, static IP, network rules	10% in phase 0 only
Business SME	Defines the representative model and validates numbers	4 hours per week during PoC
Security / compliance	Reviews the identity model	2 sessions

The Datasphere administrator is the single point of failure in this plan. If that person is not assigned by day 3, escalate.

8. Risks

Risk	Likelihood	Impact	Mitigation
Datasphere admin unavailable, PoC blocked	High	High	Name the person and secure a time commitment before kickoff
Semantic layer duplicated in both platforms	High	High	Decide the authoritative layer in phase 3 and document it as a binding decision
Security model rejected in review after the build	Medium	High	Bring security into phase 0, not phase 4
Datasphere consumption cost exceeds expectation	Medium	Medium	Measure capacity units in the PoC and model the cost before commitment
BDC Connect for Fabric slips its date	Medium	Medium	Do not architect on an unreleased capability; design an interim path that migrates cleanly
Gateway VM becomes an unmanaged single point of failure	Medium	Medium	Gateway cluster, monitoring, documented ownership from day 1
DirectQuery performance disappoints in UAT	Medium	Medium	Measure p95 in the PoC on the real model, not a sample

Feasibility: high. Multiple proven paths exist and at least two require no additional licensing. The risk is not technical failure — it is choosing a path that cannot satisfy the security requirement, and discovering that late.

9. Expected results

1. Current-state landscape diagram
2. Options analysis with a weighted scoring matrix against the customer's own criteria
3. Measured PoC results — performance, cost, semantic fidelity, security behavior
4. Target architecture diagram (logical and physical) with an architecture decision record
5. Security and identity design
6. Cost model covering licensing, infrastructure, and Datasphere consumption
7. Implementation roadmap with phases, effort, and dependencies
8. Risk register
9. Operations runbook outline for the chosen path

10. Open questions for the customer

1. Is this plain SAP Datasphere, or SAP Business Data Cloud? (*This single answer eliminates or unlocks Option 5.*)
2. Do business semantics — hierarchies, currency conversion, calculated measures — currently live in Datasphere analytic models?
3. Are Datasphere Data Access Controls used for row-level security today, and is enforcement in Datasphere a compliance requirement?
4. Expected data volumes and refresh frequency per model?
5. Is SAP Analytics Cloud also in use? (*If so, the target is a hybrid front-end architecture, not a Power BI replacement.*)
6. Is there an existing on-premises data gateway, and who operates it?
7. Is Microsoft Fabric licensed, or Power BI Pro only?
8. Is there budget for a third-party connectivity product, or is the constraint "native tooling only"?
9. Who is the named Datasphere administrator for this engagement?

Questions 1, 3 and 8 determine the answer almost entirely. Get those three before anything else.

Appendix — how the two engagements connect

The governance work in the Power BI administration engagement determines *where* the Datasphere-fed semantic models live and *who owns them*. Specifically:

- Datasphere-sourced semantic models belong in **certified data workspaces**, not report workspaces.
- The gateway data source and its credentials need a named owner in the RACI.
- If the technical-user security model is chosen, the DAX RLS roles become governed artifacts requiring certification.
- The Datasphere connection is a tenant-level dependency and belongs in the tenant-settings change log.

Present them as one operating model with two workstreams. It is a stronger position than two disconnected reports.